Summary of metrics

- **Ridge** — RMSE **5.2205**, MAE **3.2901**, R^2 **0.7331**, RMSUE **4.9327**, UnderPredictionRate **0.5588**

- **Random Forest (RF)** — RMSE **4.3164**, MAE **2.4532**, R^2 **0.8175**, RMSUE **3.9169**, UnderPredictionRate **0.5392**

- **MSUE Linear** — RMSE **9.8191**, MAE **8.6326**, R^2 **0.0557**, RMSUE **1.5151**, UnderPredictionRate **0.0980**

- **MSUE Ridge** — RMSE **26.2562**, MAE **24.5454**, R^2 **−5.7517**, RMSUE **0.5008**, UnderPredictionRate **0.0588**

	Ridge	Random Forest	MSUE Linear	MSUE Ridge
RMSE	5.2205	4.3164	9.8191	26.2562
MAE	3.2901	2.4532	8.6326	24.5454
R^2	0.7331	0.8175	0.0557	-5.7517
RMSUE	4.9327	3.9169	1.5151	0.5008
Under Prediction Rate	0.5588	0.5392	0.098	0.0588

Random Forest vs Ridge Metrics

- **RMSE** decrease of **0.9041** (≈ **17.32%** reduction)

- **MAE** decrease of **0.8369** (≈ **25.44%** reduction)

- **R^2** increase of **0.0844** (≈ **11.52%** relative increase vs Ridge's R^2)

- **RMSUE** decrease of **1.0158** (≈ **20.59%** reduction)

- **UnderPredictionRate** decrease of **0.0196** (absolute drop ≈ **1.96 percentage points**, ≈ **3.51%** relative)
- **Summary:** RF is objectively better than Ridge across typical metrics (RMSE, MAE, R^2) and also reduces underprediction magnitude moderately.

MSUE Linear vs Random Forest Metrics

- **RMSE: +127.5%** worse (9.82 vs 4.32)

- **MAE: +251.9%** worse (8.63 vs 2.45)

- **R^2:** drops from 0.82 → 0.056 (huge loss of explanatory power)

- **RMSUE: −61.4%** improvement (1.515 vs 3.917) —much smaller underprediction magnitude

- **UnderPredictionRate: −81.8%** (0.539 → 0.098) — underpredicts far less often
- **Summary:** MSUE Linear dramatically reduces underprediction frequency and magnitude, but at the cost of much worse overall errors and almost no variance explained. That means the model is biased in a way that avoids underprediction but produces large overpredictions or very noisy predictions overall.

<u>MSUE Ridge vs Random Forest Metrics</u>

- **RMSE: +508.5%** worse (26.26 vs 4.32) — huge degradation

- **MAE: +900%** worse

- **R^2:** becomes highly negative (−5.75) — model far worse than always predicting the mean

- **RMSUE: −87.2%** improvement (0.501 vs 3.917) — tiny underprediction magnitude

- **UnderPredictionRate: −89.1%** (0.539 → 0.0588)
- **Summary:** MSUE Ridge nearly eliminates underprediction but does so by producing very large overall errors. It looks like the model learned to systematically overpredict (or produce extreme values) such that it almost never underpredicts, but the overall fit is terrible.

<u>Overall Results Summary</u>

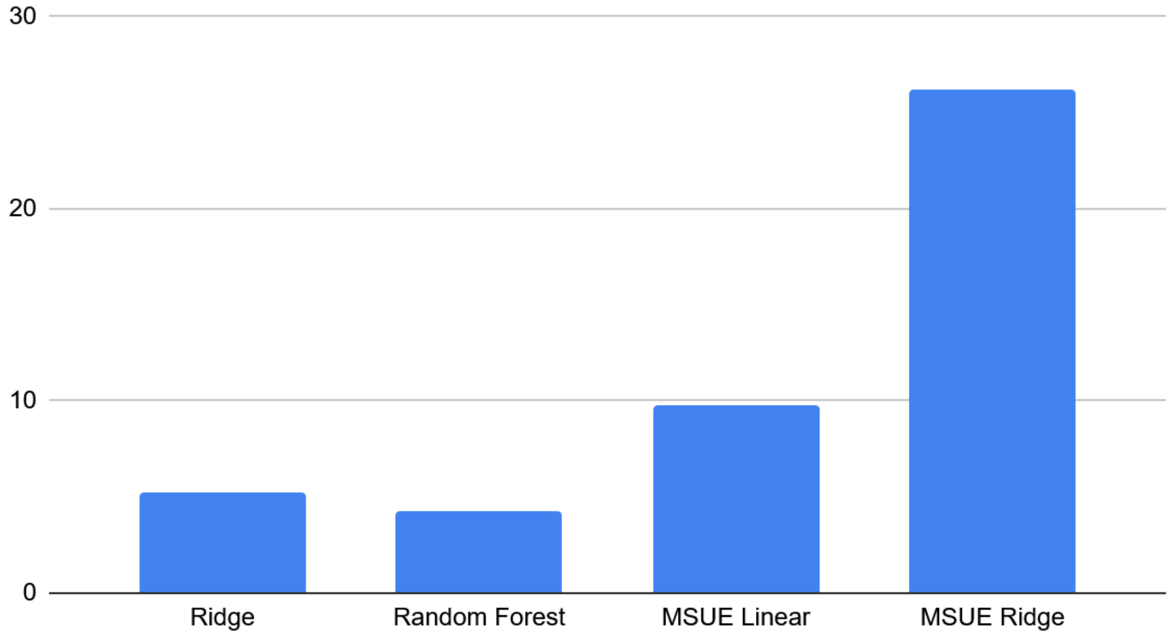
- MSUE objective is doing exactly what it's optimized for, it reduces RMSUE (underprediction magnitude) and UnderPredictionRate a lot.

- The cost of that optimization is a large increase in overprediction errors and overall error (RMSE/MAE) and loss of explanatory power (R^2). The MSUE models bias predictions upward (or to extreme values) to avoid underprediction, which hurts average accuracy drastically.

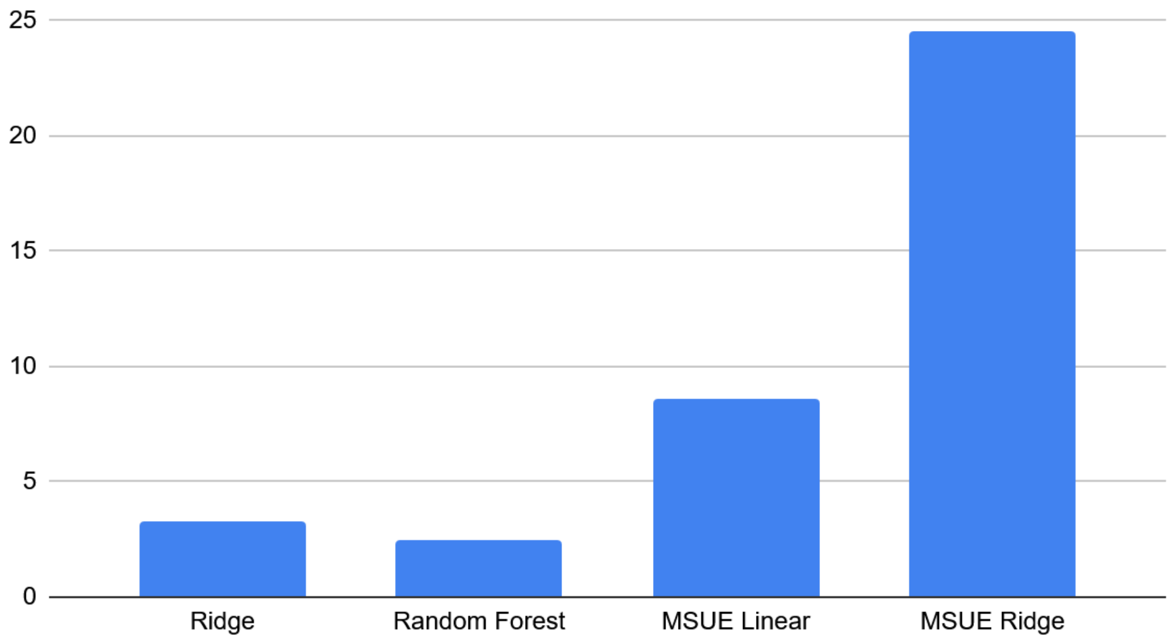
- Random Forest remains the best overall model if our goal is overall accuracy (RMSE / MAE / R^2).

- MSUE variants are only better when underprediction is *much* more detrimental than overprediction *and* our business can tolerate large overpredictions.

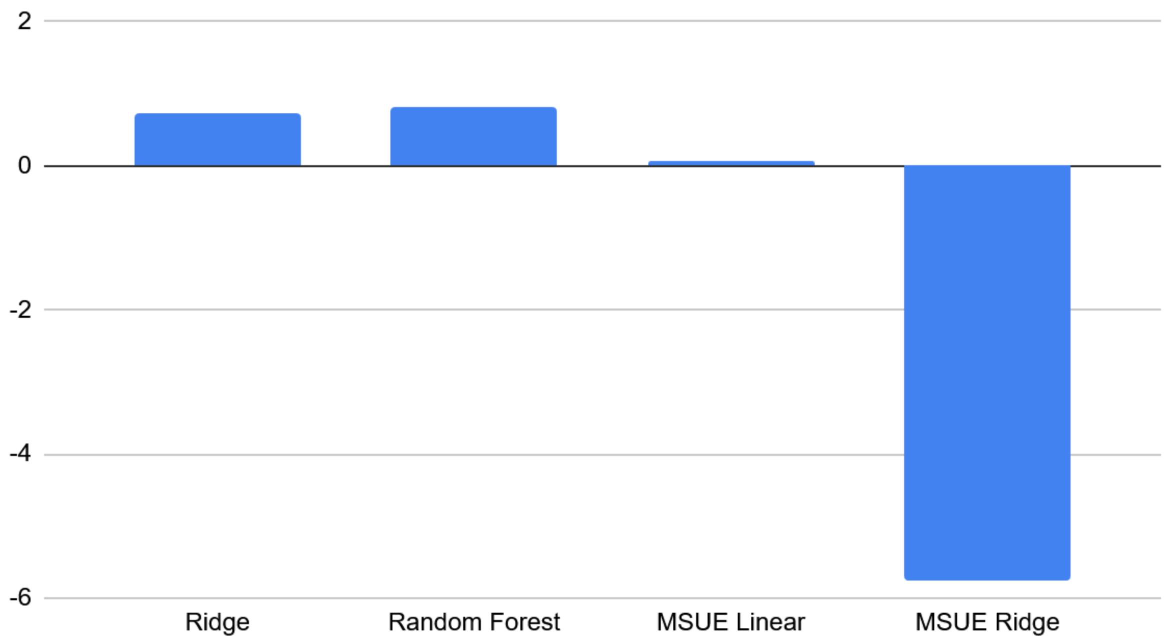
Differences in RMSE Metric Between Models



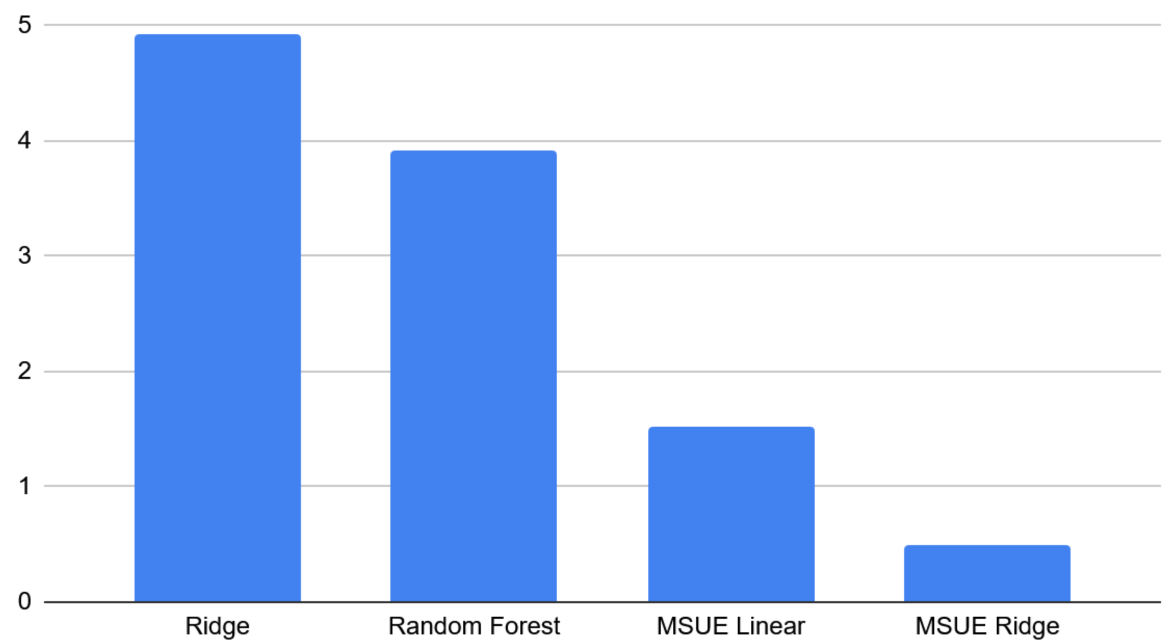
Differences in MAE Metric Between Models

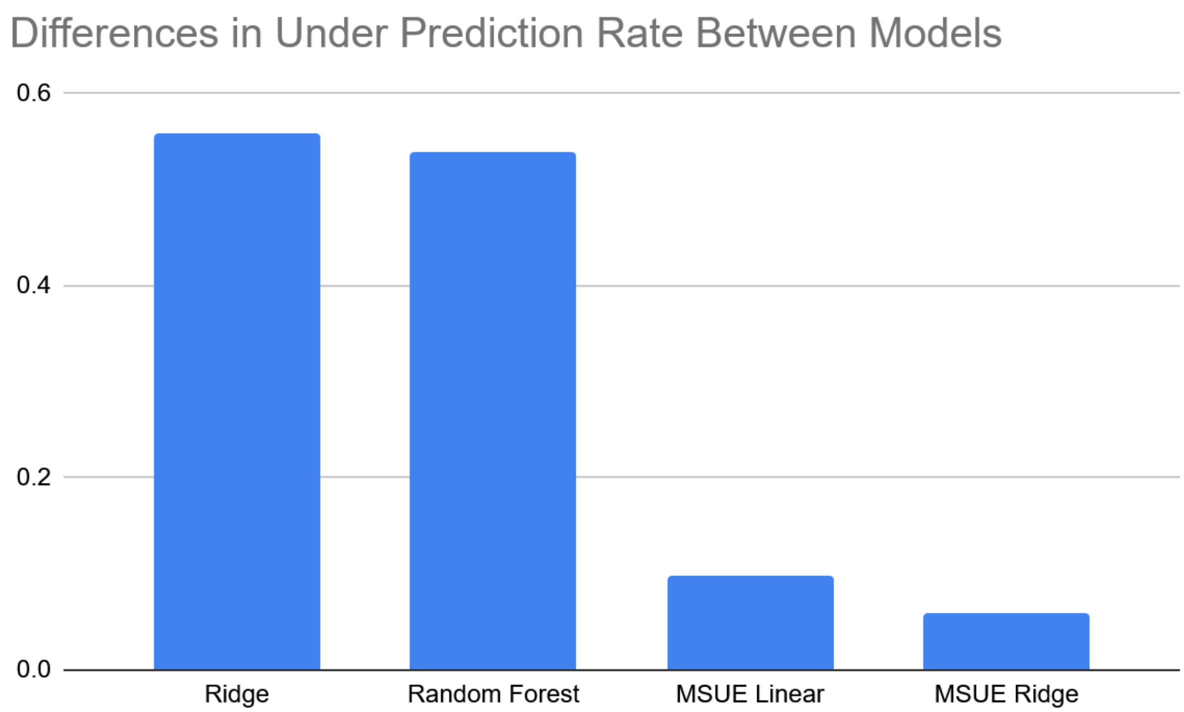


Differences in R² Metric Between Models



Differences in RMSUE Metric Between Models




Differences in Under Prediction Rate Between Models
0.6
0.4
0.2
0.0
Ridge
Random Forest
MSUE Linear
MSUE Ridge